

Assignment 6 Final Report

Model Evaluation, Explainability, and Fairness Reflection

University Advising Appointment Non-Completion Prediction

Course: MBAI 5310G - AI Programming

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Dataset: university_advising_completion_dataset.csv

Model Used: Decision Tree Classifier

Business Target: appointment_not_completed

Positive Class: Not Completed

This report evaluates a classification model, explains its predictions using SHAP and LIME, analyzes model errors, and reflects on fairness across student groups.

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1. Executive Summary

This report presents the Assignment 6 model evaluation, explainability, and fairness reflection project using a university advising appointment dataset. The business objective is to help an advising office identify appointments that may not be completed so that the university can send timely reminders, provide follow-up support, and use advising resources more effectively.

The dataset contains 360 advising appointment records and 17 columns. The original outcome column is `appointment_completed`. For business-risk interpretation, a new target variable, `appointment_not_completed`, was created so that the positive class represents Not Completed appointments. This framing is useful because the main business concern is identifying appointments that may need intervention before the scheduled advising time.

The Decision Tree model achieved a testing accuracy of 69.44%. For the positive class, Not Completed, the model achieved precision of 54.05%, recall of 80.00%, and F1-score of 64.52%. Recall is the most important metric for this business problem because a low recall would mean the advising office misses students who may need reminders or support. The model correctly identified 20 of 25 not-completed appointments in the test set, but it also produced 17 false positives, meaning that some students who would have completed their appointments were still flagged as at risk.

The most important model features were missed appointments in the previous year, previous advising visits, current GPA, previous advisor rating, and first-year student status. SHAP and LIME were used to explain model behavior globally and locally. The fairness reflection found meaningful differences in accuracy and predicted positive rates across age group, gender, and region. Therefore, the model is useful as an early prototype and decision-support tool, but it should not be used as an automatic decision system without further validation, monitoring, and human review.

Table 1: Dataset and project summary

Item	Value
Dataset file	university_advising_completion_dataset.csv
Original dataset shape	360 rows x 17 columns
Missing values	6 missing values in <code>current_gpa</code>
Duplicate records	0 duplicate records
Original target variable	<code>appointment_completed</code>
Business target used	<code>appointment_not_completed</code>
Positive class	Not Completed
Problem type	Supervised binary classification
Model used	DecisionTreeClassifier

2. Introduction and Assignment Objective

The objective of Assignment 6 is to evaluate a machine learning classification model, interpret its predictions, analyze its errors, and reflect on fairness and bias. The assignment goes beyond basic accuracy reporting by requiring model evaluation, confusion matrix interpretation, error analysis, cross-validation,

overfitting review, feature importance, SHAP explanation, LIME explanation, and fairness analysis across group-related columns.

This report follows the same professional structure used in previous assignments, but it adds deeper model evaluation and explainability. The project is organized around a university advising context where the model predicts whether an appointment is likely to be completed or not completed. The report connects technical results to practical advising decisions, responsible AI use, and student-support considerations.

3. Business Context and Machine Learning Decision

University advising offices support students with course planning, academic risk issues, graduation checks, career guidance, and program-related decisions. Missed advising appointments can reduce access to timely support, create unused appointment capacity, and delay important academic planning. If the advising office can identify appointments that are at risk of not being completed, staff can send reminders, offer alternative appointment modes, or provide targeted follow-up before the appointment date.

The machine learning decision supported by the model is: Which advising appointments should be prioritized for additional reminder or support action because they are at risk of not being completed?

Table 2: Business challenge and possible advising action

Business Challenge	Why It Matters	Possible Business Action
Appointment non-completion	Missed appointments reduce student access to advising and waste scheduled advising capacity.	Send reminders, confirm attendance, or offer rescheduling options.
Limited advising resources	Staff time should be directed toward students who are most likely to need additional support.	Use predictions to prioritize outreach rather than contacting every student equally.
Academic-support risk	Students who miss advising may delay course planning, academic-risk intervention, or graduation preparation.	Provide proactive support for appointments predicted as not completed.
Equity and fairness	Model outputs should not systematically disadvantage particular age, gender, or regional groups.	Review fairness metrics and apply human judgment before acting on predictions.

4. Dataset Overview

The dataset used for this project is `university_advising_completion_dataset.csv`. Each row represents one advising appointment. The dataset includes student profile information, advising context, appointment booking behavior, previous advising behavior, academic indicators, reminder information, appointment mode, previous advising rating, and appointment completion outcome.

The dataset has 360 rows and 17 columns. There are no duplicate records. The only column with missing values is `current_gpa`, which has 6 missing values. The original target variable `appointment_completed` contains 233 completed appointments and 127 not-completed appointments. After converting the outcome to `appointment_not_completed`, the positive class represents the business-risk outcome: Not Completed.

Table 3: Dataset columns and business meaning

Column	Role	Business Meaning
appointment_id	Identifier - removed	Unique appointment code.
student_age	Age proxy - removed	Student age in years; removed to reduce age proxy risk.
age_group	Fairness only	Age group used only for fairness reflection.
gender	Fairness only	Gender group used only for fairness reflection.
region	Fairness only	Student region used only for fairness reflection.
student_level	Model input	Academic level of the student.
program_area	Model input	Student program or discipline area.
appointment_type	Model input	Type of advising appointment.
booking_lead_days	Model input	Days between booking and appointment.
previous_advising_visits	Model input	Number of previous advising visits.
missed_appointments_last_year	Model input	Missed advising appointments in the previous year.
credits_completed	Model input	Credits already completed by the student.
current_gpa	Model input	Current GPA; missing values imputed with median.
email_reminder_sent	Model input	Whether an email reminder was sent.
appointment_mode	Model input	Mode of the appointment.
advisor_rating_previous	Model input	Previous advising rating.
appointment_completed	Original target	Original completion outcome.

Table 4: Missing value summary

Column	Missing Values
current_gpa	6

Table 5: Duplicate record summary

Item	Count
Duplicate records	0

Table 6: Target variable distribution after target conversion

Target Value	Meaning	Count	Percentage
0	Completed	233	64.72
1	Not Completed	127	35.28

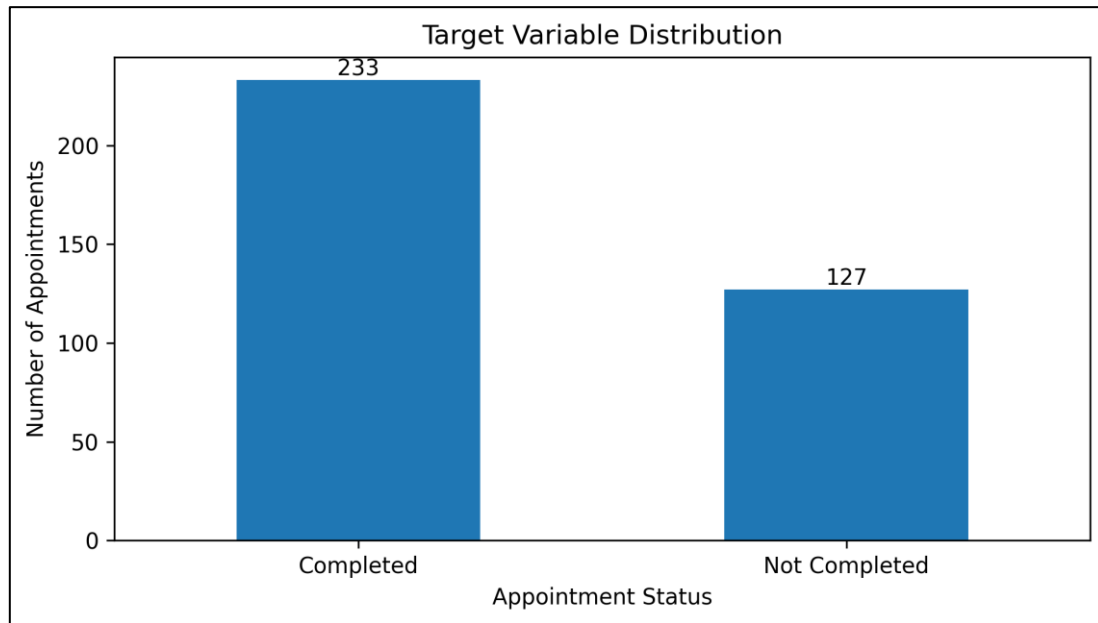


Figure 1: Target variable distribution

The target variable is moderately imbalanced because Completed appointments represent 64.72% of the dataset, while Not Completed appointments represent 35.28%. This imbalance is not extreme, but it means that accuracy alone is not sufficient. Precision, recall, F1-score, and confusion matrix values must also be interpreted.

5. Target Variable, Input Features, and Fairness Columns

The original target variable is `appointment_completed`. In the raw dataset, this column indicates whether the advising appointment was completed. For this assignment, the business target was reframed as `appointment_not_completed`, where 1 means Not Completed and 0 means Completed. This makes the positive class the risk event that the advising office wants to identify.

The model input includes numerical and categorical advising-related features. Identifier columns are removed because they do not represent meaningful student behavior. Fairness-related columns are kept outside the model and used only after prediction to compare model behavior across groups. The `student_age` column is also removed because it could act as a proxy for `age_group` and may create indirect age-related discrimination risk.

Table 7: Numerical and categorical feature groups used for modelling

Feature Type	Number of Features	Columns
Numerical	6	booking_lead_days, previous_advising_visits, missed_appointments_last_year, credits_completed, current_gpa, advisor_rating_previous
Categorical	5	student_level, program_area, appointment_type, email_reminder_sent, appointment_mode

Table 8: Removed and fairness-only columns

Column Group	Columns	Reason
Identifier column	appointment_id	Identifier does not provide meaningful predictive information.
Fairness-only columns	age_group, gender, region	Kept for fairness analysis but excluded from model input to avoid direct group-based predictions.
Proxy column removed	student_age	Removed because it may act as a proxy for age_group.
Original target removed	appointment_completed	Not used directly after creating the business-risk target.
Final target created	appointment_not_completed	Positive class = Not Completed, which is the risk the advising office wants to identify.

6. Data Preparation and Preprocessing

Data preprocessing is required because machine learning models need clean and numeric input. The dataset contains both numerical and categorical features, so different preprocessing steps are required for different feature types. Numerical missing values are handled using median imputation. Categorical variables are converted into numerical columns using one-hot encoding. These steps are combined inside a scikit-learn Pipeline with a ColumnTransformer.

Using a Pipeline is important because preprocessing is fitted only on the training data and then applied to the testing data. This helps prevent data leakage. Data leakage occurs when information from the testing set influences model training, which can make performance look better than it really is.

Table 9: Preprocessing approach

Feature Group	Columns	Preprocessing Applied
Numerical features	booking_lead_days, previous_advising_visits, missed_appointments_last_year, credits_completed, current_gpa, advisor_rating_previous	SimpleImputer(strategy="median")
Categorical features	student_level, program_area, appointment_type, email_reminder_sent, appointment_mode	SimpleImputer(strategy="most_frequent") and OneHotEncoder(handle_unknown="ignore")

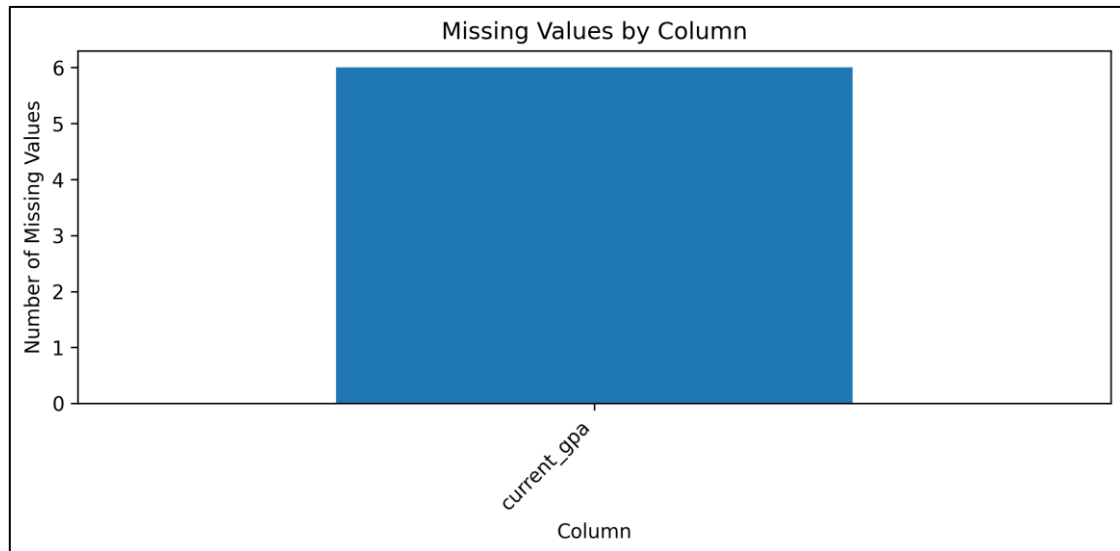


Figure 2: Missing values by column before preprocessing

7. Train-Test Split and Decision Tree Modelling

The dataset was divided into training and testing sets using an 80/20 split with `random_state=42` and `stratify=y`. The training data is used to teach the model patterns from historical appointments. The testing data is kept separate and used to evaluate how well the trained model performs on unseen appointments.

A Decision Tree Classifier was selected because the assignment requires a simple Decision Tree classification model. The tree was controlled using `max_depth=3` and `min_samples_leaf=8` so that the model remains interpretable and does not become too complex.

Table 10: Train-test split summary

Dataset	Shape
X_train	(288, 11)
X_test	(72, 11)
y_train	(288,)
y_test	(72,)

Table 11: Target class distribution in training and testing sets

appointment_not_completed	Training Set	Testing Set
Completed	186	47
Not Completed	102	25

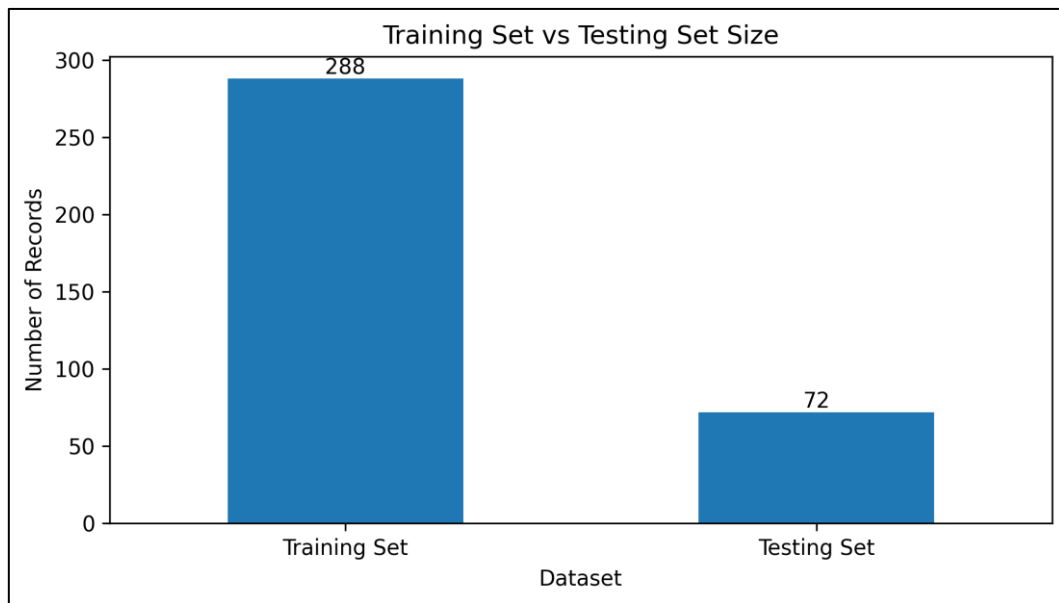


Figure 3: Training set vs testing set size

Table 12: Sample actual and predicted values

appointment_id	Actual Meaning	Predicted Meaning	Predicted Probability - Not Completed	Correct Prediction
UA0077	Completed	Completed	0.0777	Correct
UA0045	Not Completed	Not Completed	0.5119	Correct
UA0137	Not Completed	Not Completed	0.5119	Correct
UA0263	Not Completed	Not Completed	0.5119	Correct
UA0308	Completed	Completed	0.0777	Correct
UA0283	Not Completed	Not Completed	0.8846	Correct
UA0116	Not Completed	Not Completed	0.8846	Correct
UA0109	Completed	Completed	0.0777	Correct
UA0056	Not Completed	Not Completed	0.8846	Correct
UA0233	Not Completed	Completed	0.0777	Incorrect
UA0252	Not Completed	Not Completed	0.5119	Correct
UA0360	Completed	Completed	0.3846	Correct

The sample actual-versus-predicted table shows that the model correctly classified many of the first test records, but it also made some mistakes. These individual comparisons are useful because they connect model predictions back to real appointment-level decisions.

8. Model Evaluation Results

The model was evaluated using accuracy, precision, recall, and F1-score. Accuracy measures the overall percentage of correct predictions. Precision for Not Completed measures how many appointments predicted as Not Completed were actually not completed. Recall for Not Completed measures how many actual Not Completed appointments the model successfully found. F1-score balances precision and recall into one metric.

The model achieved 69.44% testing accuracy. For the Not Completed class, precision was 54.05%, recall was 80.00%, and F1-score was 64.52%. For this advising problem, recall for Not Completed is the most important metric because missing an appointment that will not be completed may prevent the advising office from intervening in time.

Table 13: Decision Tree evaluation metrics

Metric	Score	Percentage
Accuracy	0.6944	69.44%
Precision for Not Completed	0.5405	54.05%
Recall for Not Completed	0.8000	80.00%
F1-score for Not Completed	0.6452	64.52%

Table 14: Classification report

Class/Metric	precision	recall	f1-score	support
Completed	0.8571	0.6383	0.7317	47
Not Completed	0.5405	0.8000	0.6452	25
Overall accuracy			0.6944	
macro avg	0.6988	0.7191	0.6884	72
weighted avg	0.7472	0.6944	0.7017	72

Note: For the overall accuracy row, precision, recall, and support are not applicable. The value shown is the overall test accuracy.

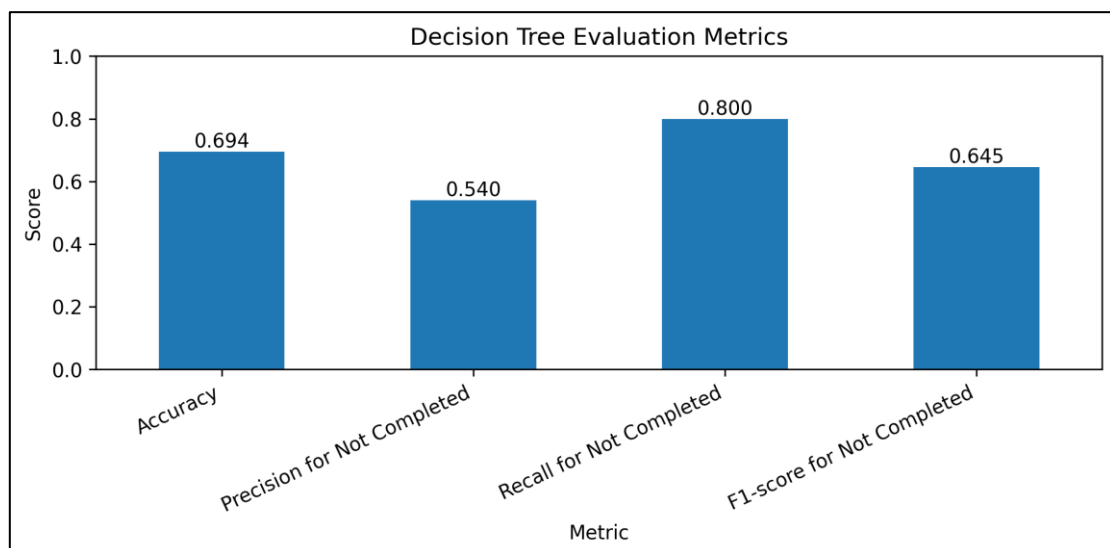


Figure 4: Decision Tree evaluation metrics

9. Confusion Matrix and Business Error Analysis

The confusion matrix provides a deeper view of model performance by separating correct and incorrect predictions. In this report, the positive class is Not Completed. Therefore, a true positive means that the model correctly identified an appointment that was not completed.

Table 15: Confusion matrix values

Unnamed: 0	Predicted Completed	Predicted Not Completed
Actual Completed	30	17
Actual Not Completed	5	20

Table 16: Confusion matrix business meaning

Item	Value	Business Meaning
True Negative	30	Completed appointments correctly predicted as completed.
False Positive	17	Completed appointments incorrectly predicted as not completed; may cause unnecessary reminders or support.
False Negative	5	Not completed appointments incorrectly predicted as completed; may cause missed intervention opportunities.
True Positive	20	Not completed appointments correctly identified for possible support.

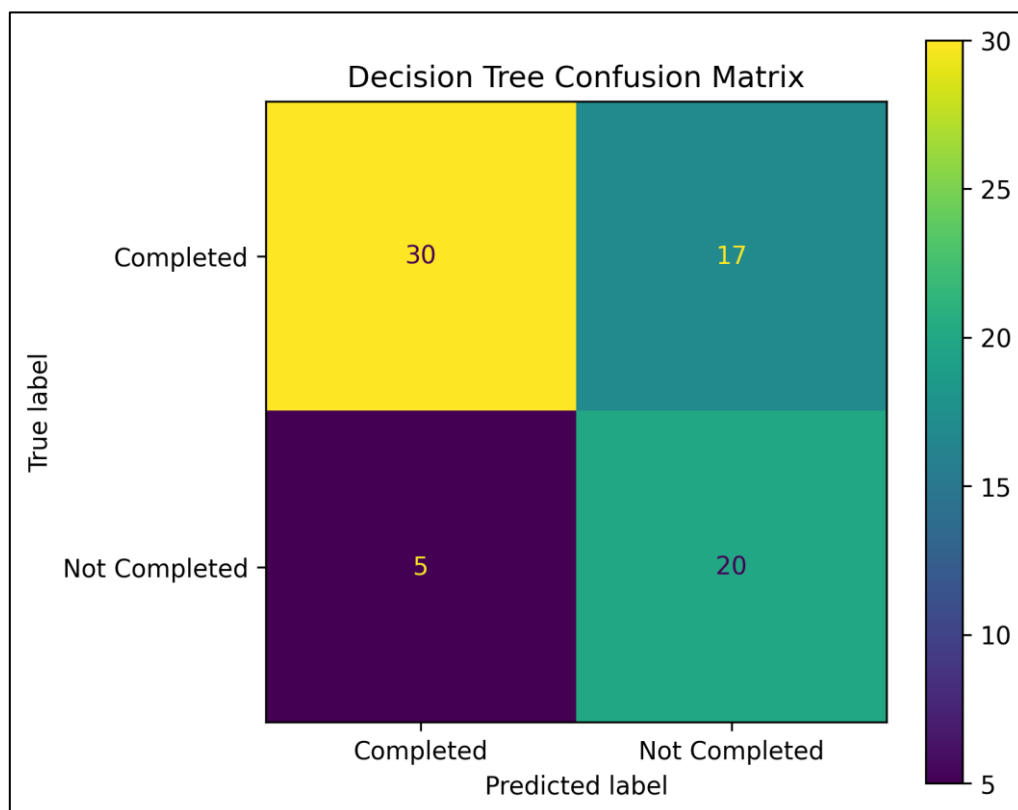


Figure 5: Decision Tree confusion matrix

The model produced 30 true negatives and 20 true positives. It also produced 17 false positives and 5 false negatives. A false negative is especially serious because the model predicts that the appointment will be completed when it is actually not completed. This may cause the advising office to miss an opportunity to send reminders or provide support. False positives also matter because they may create unnecessary reminders or staff follow-up, but their business cost is usually less serious than missing a student who needs intervention.

10. Error Analysis

Error analysis examines the individual records where the model is wrong. This helps identify whether the model is making one type of mistake more often than another and whether errors appear to cluster around certain feature values or student groups.

Table 17: Error type counts

Error Type	Count
True Negative	30
True Positive	20
False Positive	17
False Negative	5

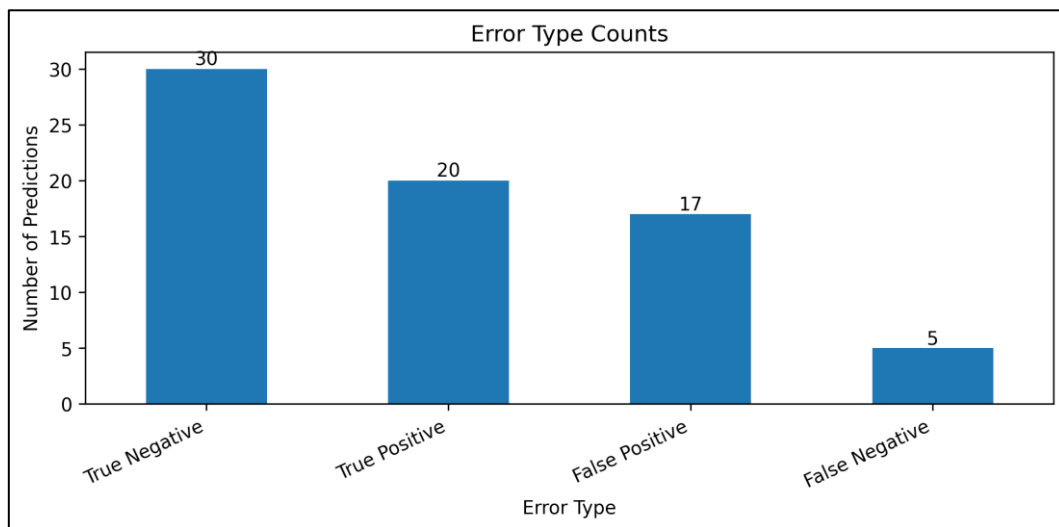


Figure 6: Error type counts

Table 18: Sample wrong predictions

appointment_id	Actual Meaning	Predicted Meaning	Error Type	age_group	gender	region	student_level	Missed_ appointments last_year	Previous Advising_ visits	current_gpa
UA0233	Not Completed	Completed	False Negative	26-35	Female	South	Graduate	0	7	2.70
UA0019	Completed	Not Completed	False Positive	18-25	Female	East	First Year	2	8	3.20
UA0170	Completed	Not Completed	False Positive	36-50	Male	West	Third Year	2	4	2.67
UA0318	Completed	Not Completed	False Positive	36-50	Male	East	Third Year	0	1	2.96
UA0081	Completed	Not Completed	False Positive	18-25	Male	West	Second Year	0	6	2.07
UA0002	Completed	Not Completed	False Positive	26-35	Female	East	First Year	0	6	2.94
UA0065	Not Completed	Completed	False Negative	26-35	Male	West	Second Year	0	9	2.08
UA0302	Completed	Not Completed	False Positive	26-35	Female	South	First Year	2	9	3.48
UA0148	Completed	Not Completed	False Positive	36-50	Male	West	First Year	1	5	1.92
UA0242	Completed	Not Completed	False Positive	36-50	Male	West	Second Year	2	5	2.81

The most common error type is false positive, with 17 cases. In these cases, the model predicted Not Completed even though the appointment was actually completed. This means the model is cautious and tends to flag more appointments as at risk. This behavior supports high recall, but it also creates extra outreach work. The false negative count is lower, with 5 cases, which is favorable for a support-focused advising context because fewer actual non-completions are missed.

The wrong-prediction sample suggests that errors often occur when students have mixed signals. Several false positives appear in the 26-35 and 36-50 age groups, and many of these records still show risk-related signals such as low GPA, previous advising visits, career or graduation-check appointment types, or previous missed appointments. However, some false positives also have 0 missed appointments in the previous year, which means the model sometimes flags students as at risk even when one strong protective signal is present. This pattern helps explain why the model achieves high recall but lower precision: it is designed to catch many possible non-completions, but it may over-identify risk for students who would still attend.

This pattern-level error analysis connects directly to the fairness reflection. If false positives are concentrated in particular age groups, regions, or advising contexts, the university should avoid treating the prediction as a final decision. The safer use is to treat flagged appointments as candidates for helpful reminders or optional support, not as evidence that a student is irresponsible or unlikely to attend.

11. Cross-Validation and Model Stability

Cross-validation was used to evaluate whether the model performance is stable across different subsets of the data. A 5-fold stratified cross-validation approach was used. The fold accuracies ranged from 61.11% to 72.22%. The mean cross-validation accuracy was 66.67%, with a standard deviation of 4.21 percentage points.

Table 19: Cross-validation accuracy results

Fold	Accuracy	Accuracy Percentage
Fold 1	0.6250	62.50%
Fold 2	0.7222	72.22%
Fold 3	0.6111	61.11%
Fold 4	0.6944	69.44%
Fold 5	0.6806	68.06%

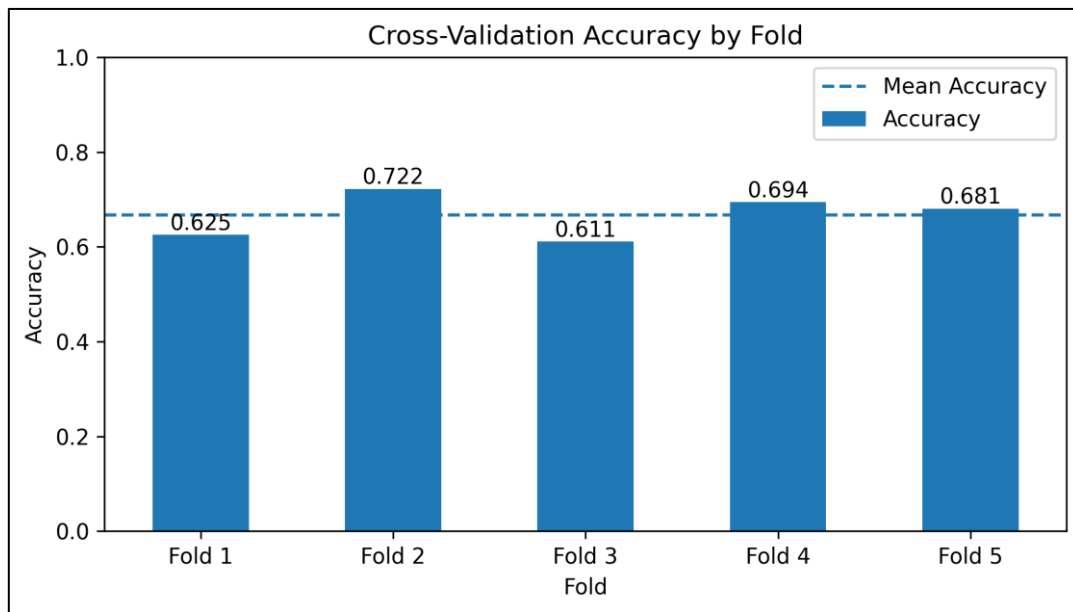


Figure 7: Cross-validation accuracy by fold

The cross-validation results show moderate variation across folds. The lowest fold accuracy is 61.11% and the highest fold accuracy is 72.22%, creating an 11.11 percentage-point spread. This difference is meaningful because the full dataset has only 360 records, so each fold contains approximately 72 records. With small fold sizes, a few changed predictions can noticeably shift the fold accuracy. Therefore, the model has useful predictive signal, but its stability should be interpreted cautiously and tested again with a larger advising dataset before operational use.

12. Overfitting and Underfitting Analysis

Overfitting occurs when a model learns the training data too closely and performs much worse on new testing data. Underfitting occurs when a model is too simple to capture useful patterns in either the training or testing data. To assess this, training accuracy and testing accuracy were compared, and model performance was reviewed across tree depths from 1 to 10.

Table 20: Training and testing accuracy comparison

Evaluation Set	Accuracy	Percentage
Training Set	0.7431	74.31%
Testing Set	0.6944	69.44%
Difference	0.0486	4.86%

Table 21: Tree depth comparison

Tree Depth	Training Accuracy	Testing Accuracy	Accuracy Gap
1	0.7292	0.6806	0.0486
2	0.7361	0.6806	0.0556
3	0.7431	0.6944	0.0486
4	0.8090	0.5833	0.2257

Tree Depth	Training Accuracy	Testing Accuracy	Accuracy Gap
5	0.8160	0.6389	0.1771
6	0.8299	0.6667	0.1632
7	0.8368	0.6250	0.2118
8	0.8333	0.6944	0.1389
9	0.8333	0.6944	0.1389
10	0.8333	0.6944	0.1389

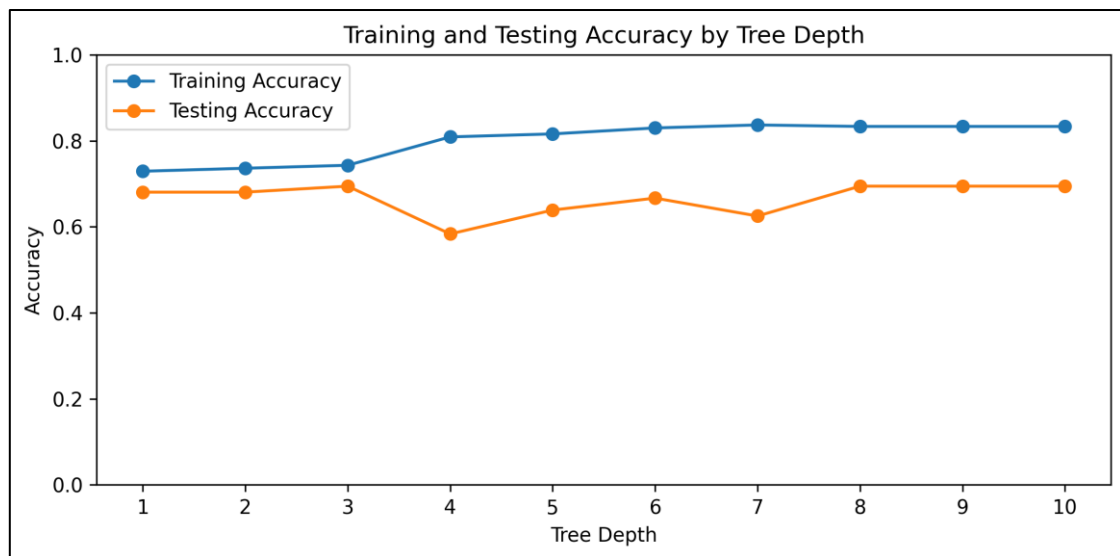


Figure 8: Training and testing accuracy by tree depth

The selected model achieved 74.31% training accuracy and 69.44% testing accuracy, producing a gap of 4.86 percentage points. This gap is relatively small and does not show strong overfitting. However, the depth comparison shows that deeper trees improve training accuracy but can reduce testing accuracy and increase the accuracy gap. This supports the decision to use a controlled-depth tree for interpretability and generalization.

13. Feature Importance

Decision Trees provide feature importance values that indicate which features contributed most to model splits. Feature importance helps translate the model into business language by showing which advising and student-context variables influenced predictions. However, feature importance does not prove cause and effect. A feature can be useful for prediction without directly causing appointment non-completion.

Table 22: Top 10 feature importance values

Feature	Importance	Importance Percentage
missed_appointments_last_year	0.445181	44.52%
previous_advising_visits	0.243925	24.39%
current_gpa	0.189158	18.92%
advisor_rating_previous	0.096745	9.67%

Feature	Importance	Importance Percentage
student_level_First Year	0.024991	2.50%
booking_lead_days	0.000000	0.00%
credits_completed	0.000000	0.00%
student_level_Fourth Year	0.000000	0.00%
student_level_Graduate	0.000000	0.00%
student_level_Second Year	0.000000	0.00%

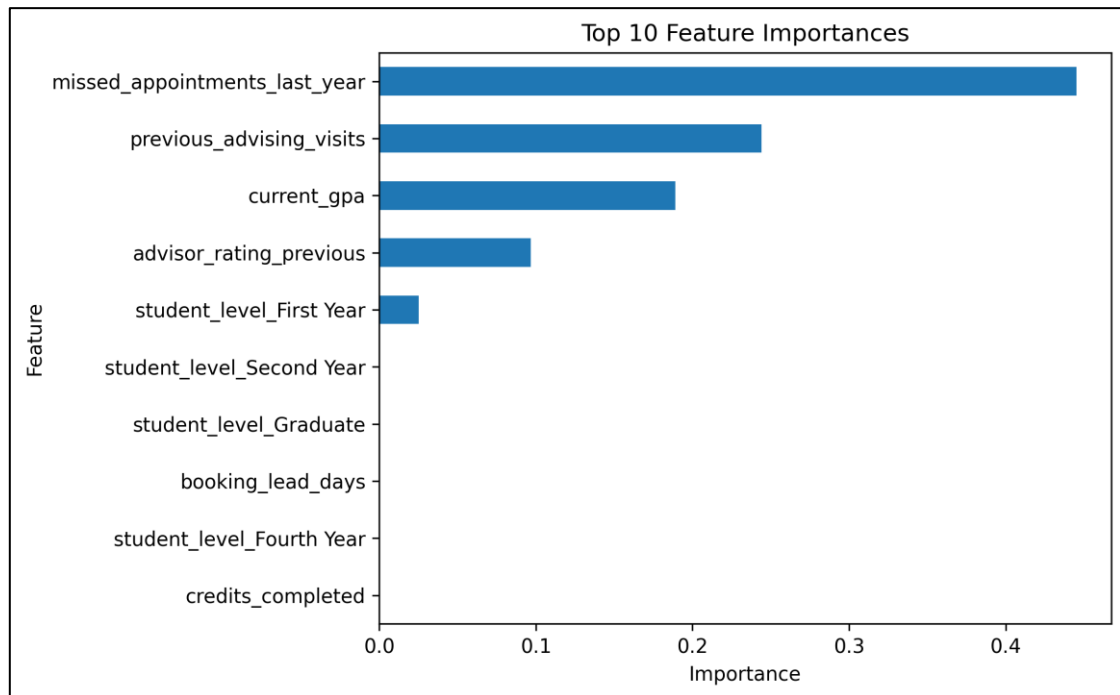


Figure 9: Top 10 feature importances

The strongest feature is `missed_appointments_last_year`, with 44.52% importance. This suggests that previous missed appointments are highly informative for identifying future appointment non-completion risk. Previous advising visits, current GPA, advisor rating, and first-year student status also contribute to the model. From a business perspective, these features suggest that both advising history and academic context may help identify appointments requiring additional support.

14. SHAP Explanation

SHAP explains how each feature contributes to model predictions. Unlike basic feature importance, SHAP can show both the size and direction of each feature effect. A positive SHAP value pushes the model toward the Not Completed prediction, while a negative SHAP value pushes the model toward the Completed prediction.

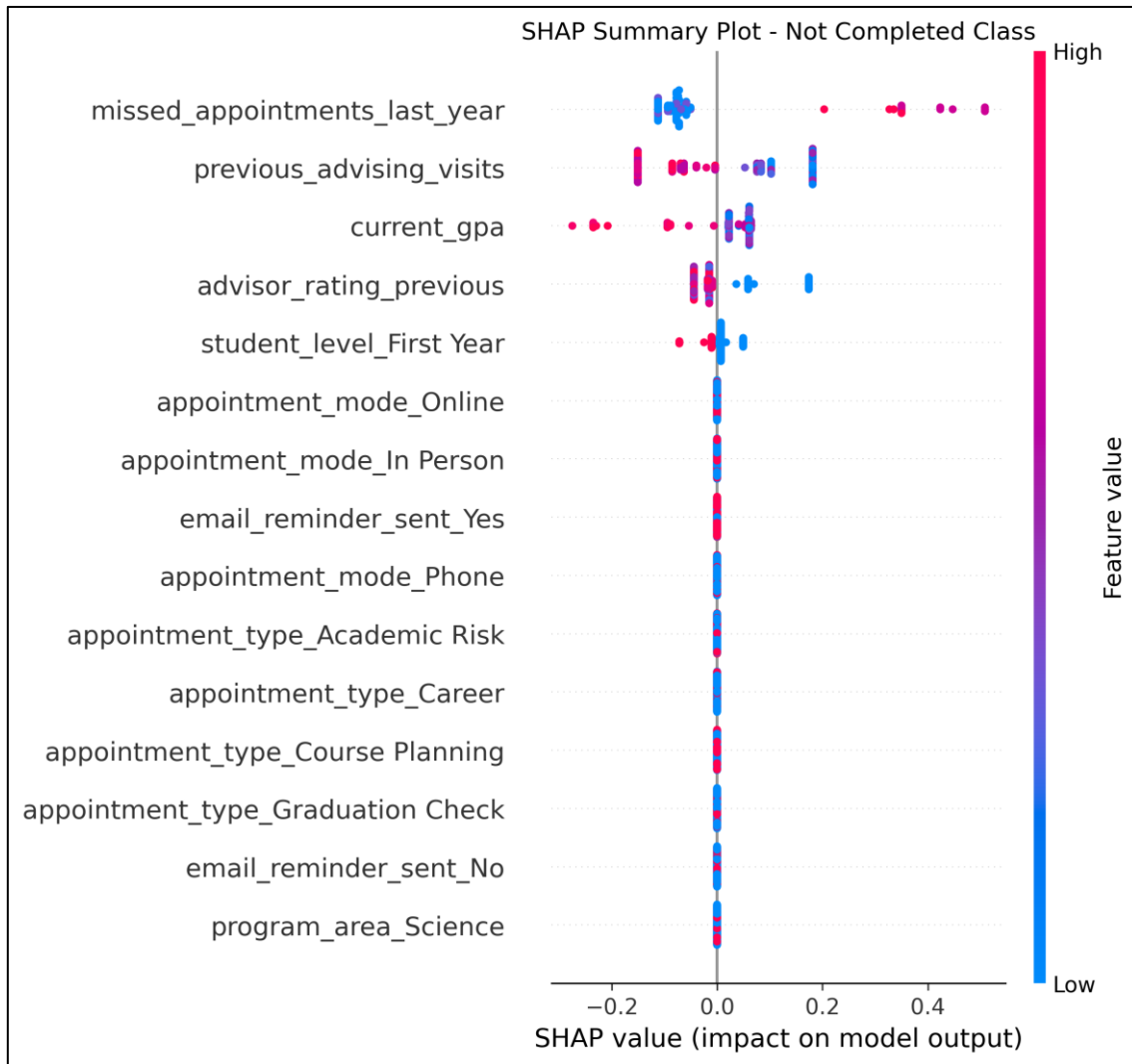


Figure 10: SHAP summary plot

The SHAP summary confirms that missed_appointments_last_year, previous_advising_visits, current_gpa, advisor_rating_previous, and student_level_First Year are the most influential features. In the global SHAP pattern, having previous missed appointments generally pushes predictions toward Not Completed. Lower current GPA and lower previous advisor rating also tend to increase Not Completed risk. In contrast, no previous missed appointments, stronger advising history, higher GPA, and stronger previous advisor rating tend to reduce the predicted risk. These patterns are consistent with the business logic that students with weaker academic or advising engagement signals may need more support.

Table 23: SHAP explanation for one selected record - top features

Feature	Feature Value	SHAP Value for Not Completed	Absolute SHAP Value
previous_advising_visits	6.00	0.181343	0.181343
current_gpa	1.70	0.060126	0.060126
missed_appointments_last_year	0.00	-0.058424	0.058424
advisor_rating_previous	2.00	-0.014873	0.014873
student_level_First Year	1.00	-0.010434	0.010434

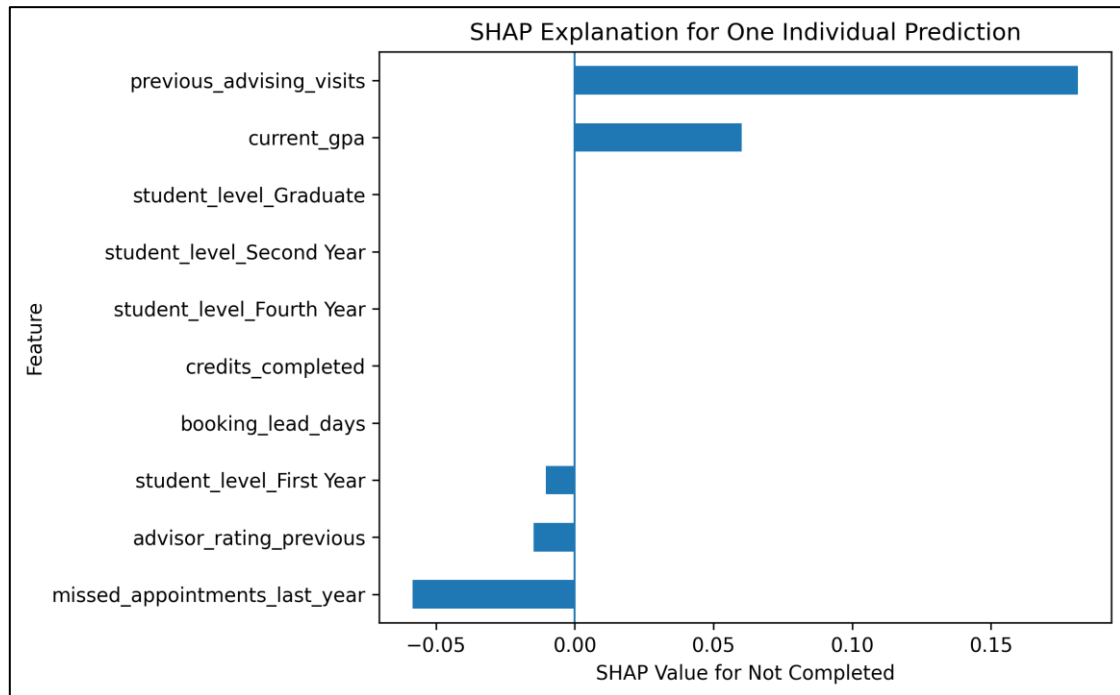


Figure 11: SHAP explanation for one individual record

For the selected individual record, the SHAP table provides a clear case-level explanation. The feature `previous_advising_visits` has the strongest positive SHAP value, so it pushes the model toward the Not Completed class for this student. The `current_gpa` value also has a positive SHAP value, meaning that the GPA level in this record increases the predicted non-completion risk. In contrast, `missed_appointments_last_year` has a negative SHAP value because this student had 0 missed appointments in the previous year. In business terms, that is a protective signal because the student has not recently missed advising appointments.

The smaller negative SHAP values for `advisor_rating_previous` and `student_level_First Year` also push the prediction slightly toward Completed. Therefore, the selected student has both risk-increasing and risk-reducing signals. SHAP is useful here because it explains why the model should not be interpreted as a simple rule. The final prediction is the result of competing evidence: advising history and GPA increase concern, while no previous missed appointment reduces concern. This is exactly the type of case where human advising judgment should review the prediction before any action is taken.

15. LIME Explanation

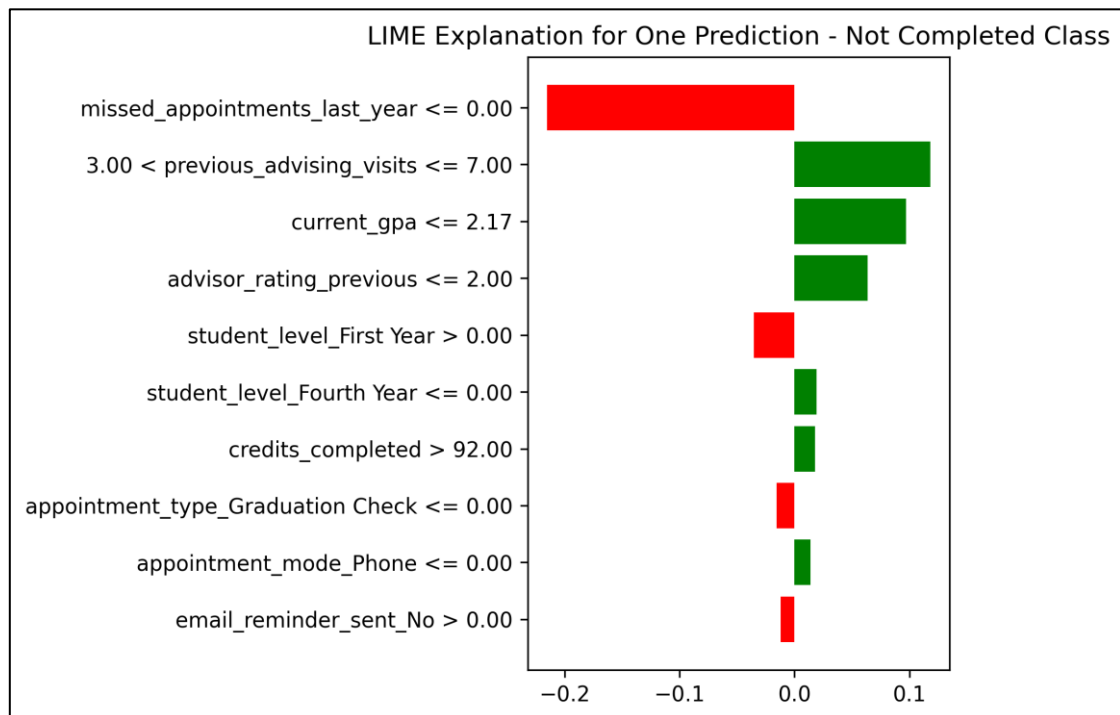
LIME explains one individual prediction by approximating the model locally around that record. This helps business users understand why the model made a particular prediction for a specific appointment. While SHAP provides both global and local explanations, LIME is especially useful for explaining one case in simple feature-condition language.

Table 24: Prediction probability for selected LIME record

Class	Predicted Probability
Completed	0.4881
Not Completed	0.5119

Table 25: LIME explanation table for one selected record

Feature Condition	Contribution to Not Completed Prediction
missed_appointments_last_year <= 0.00	-0.215532
3.00 < previous_advising_visits <= 7.00	0.118254
current_gpa <= 2.17	0.097072
advisor_rating_previous <= 2.00	0.063448
student_level_First Year > 0.00	-0.035184
student_level_Fourth Year <= 0.00	0.019260
credits_completed > 92.00	0.017948
appointment_type_Graduation Check <= 0.00	-0.015528
appointment_mode_Phone <= 0.00	0.013901
email_reminder_sent_No > 0.00	-0.012217

*Figure 12: LIME explanation for one individual prediction*

For the selected record, the predicted probability of Not Completed is 51.19%, while the predicted probability of Completed is 48.81%. This means the model is only slightly more confident in the Not Completed prediction, so the case should be treated carefully rather than automatically.

The LIME explanation shows that `current_gpa <= 2.17` and `advisor_rating_previous <= 2.00` support the Not Completed prediction. The condition `3.00 < previous_advising_visits <= 7.00` also supports Not Completed for this local decision. However, `missed_appointments_last_year <= 0.00` strongly works against the Not Completed prediction, suggesting that no recent missed appointments is a protective signal.

This is a useful business explanation because it separates risk-increasing conditions from risk-reducing conditions for one specific appointment.

SHAP and LIME do not need to rank every feature in exactly the same way because they explain the model through different approaches. SHAP distributes prediction influence across features using contribution values, while LIME builds a local approximation around the selected record. Therefore, the difference between SHAP emphasizing previous_advising_visits and LIME emphasizing missed_appointments_last_year ≤ 0.00 should be understood as complementary evidence rather than a contradiction.

16. Fairness and Bias Reflection

Fairness analysis was conducted for three group-related columns: age_group, gender, and region. These columns were intentionally excluded from model training and used only after predictions were made. For each group, the analysis reports the number of test records, actual positive rate, predicted positive rate, and accuracy by group.

Table 26: Fairness summary by age group

Age Group	Records	Actual Not Completed Rate	Predicted Not Completed Rate	Accuracy
18-25	23	39.13%	47.83%	82.61%
26-35	19	31.58%	52.63%	57.89%
36-50	30	33.33%	53.33%	66.67%

Table 27: Fairness summary by gender

Gender	Records	Actual Not Completed Rate	Predicted Not Completed Rate	Accuracy
Female	29	27.59%	44.83%	55.17%
Male	43	39.53%	55.81%	79.07%

Table 28: Fairness summary by region

Region	Records	Actual Not Completed Rate	Predicted Not Completed Rate	Accuracy
East	15	40.00%	66.67%	60.00%
North	20	40.00%	45.00%	85.00%
South	16	25.00%	37.50%	75.00%
West	21	33.33%	57.14%	57.14%

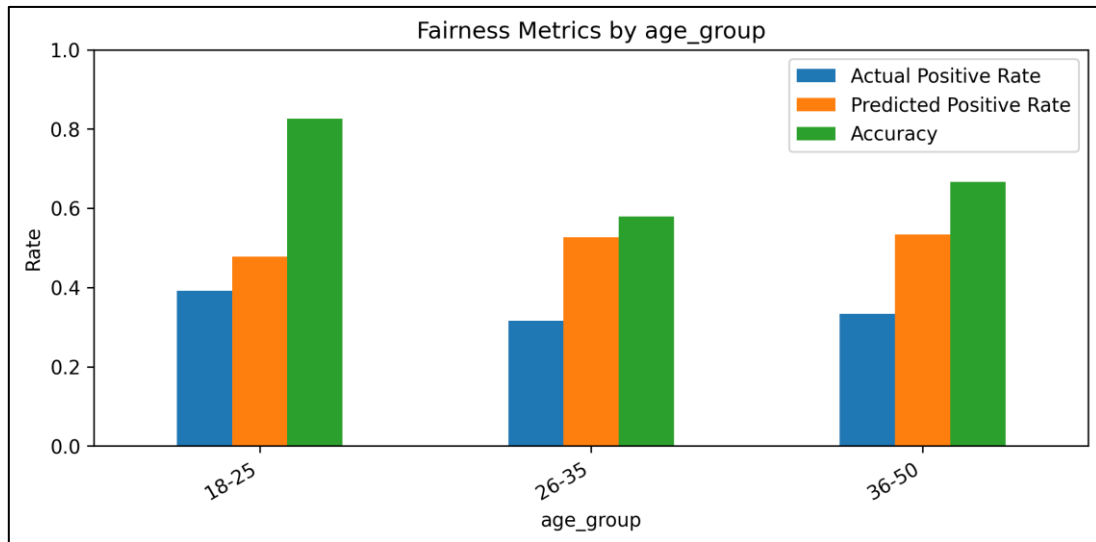


Figure 13: Fairness metrics by age group

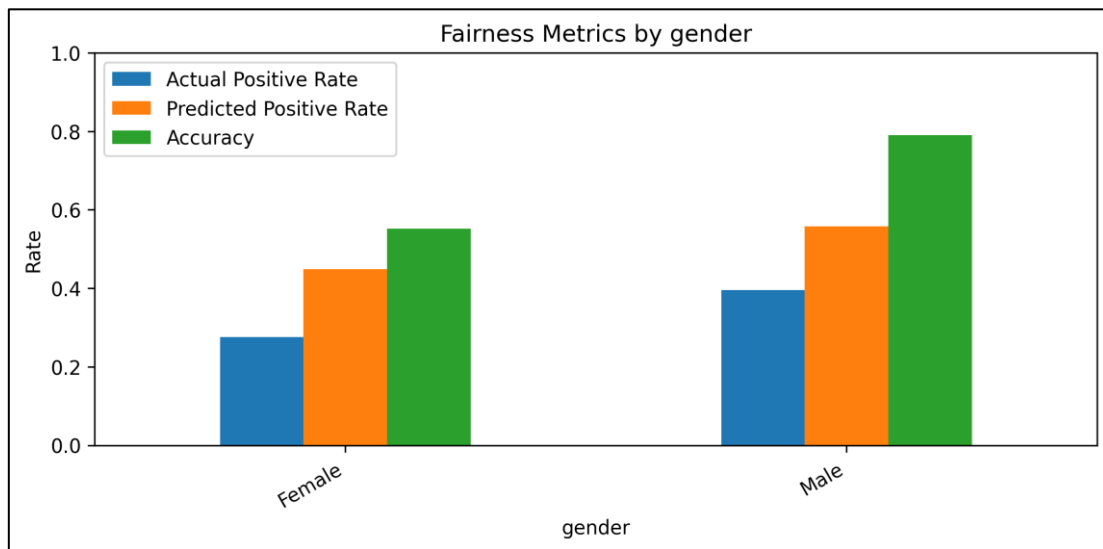


Figure 14: Fairness metrics by gender

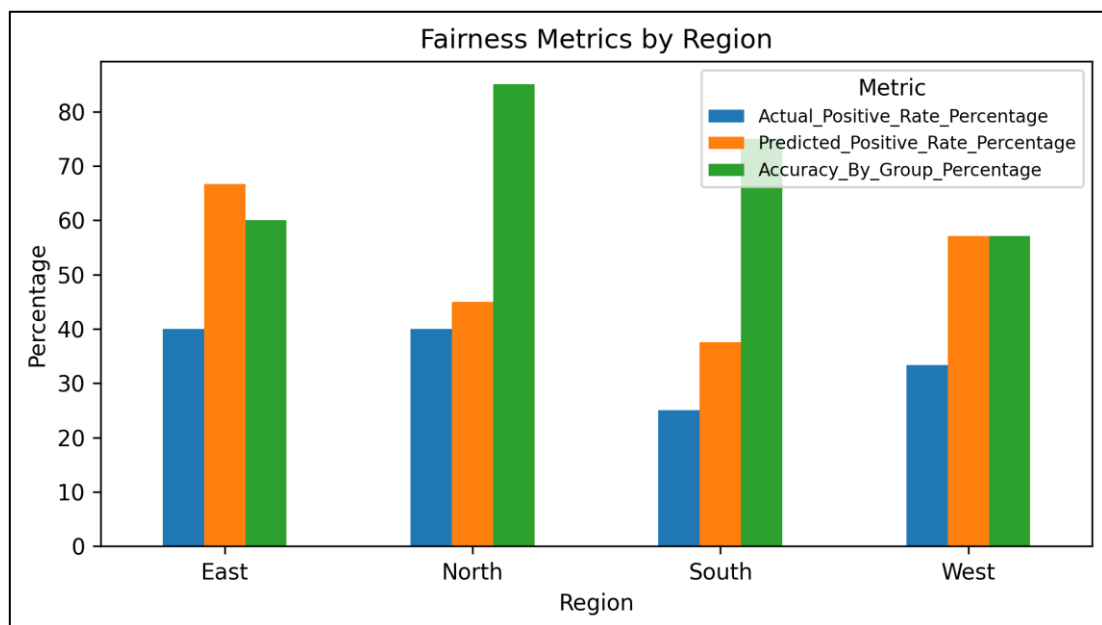


Figure 15: Fairness metrics by region

The age-group results show a noticeable accuracy gap. The 18-25 group has the highest accuracy at 82.61%, while the 26-35 group has the lowest accuracy at 57.89%. The predicted Not Completed rates are relatively close across age groups, but the accuracy difference suggests that the model may be more reliable for some age groups than others.

The gender results also show a meaningful performance gap. Accuracy is 79.07% for male students and 55.17% for female students. The predicted Not Completed rate is also higher for male students than for female students. This does not prove unfairness by itself, but it is a warning sign that the advising office should investigate whether the model is less reliable for one gender group.

The regional results show the largest differences. Accuracy is 85.00% for North, 75.00% for South, 60.00% for East, and 57.14% for West. The predicted Not Completed rate is highest in East at 66.67% and West at 57.14%, compared with 37.50% in South. These differences could affect how advising support is distributed across regions. If used without human review, the model may direct more outreach to some regions and less to others, even when actual need is not equally different.

Overall, the fairness results show that the model should not be used automatically. It may still be helpful as a screening tool, but the university should monitor group-level accuracy and predicted positive rates over time. Advising staff should review predictions before action is taken, and support should be designed as helpful outreach rather than punitive treatment.

17. Final Business Interpretation and Recommendation

The model predicts whether a university advising appointment is likely to be Not Completed. The overall model performance is useful but not strong enough for fully automated use. The model has a testing accuracy of 69.44%, but the more important result is the 80.00% recall for the Not Completed class. This means the model identifies most appointments that are actually not completed, which is valuable for proactive support.

The main error type is false positive. In business terms, the model sometimes predicts that an appointment will not be completed even though it is completed. This may cause unnecessary reminders or staff follow-up. However, the model produces only 5 false negatives, which is beneficial because false negatives represent missed opportunities to intervene before an appointment is not completed.

The overfitting analysis does not show strong overfitting for the selected model. The training-testing accuracy gap is 4.86 percentage points. Deeper trees show larger gaps, so the controlled-depth tree is a reasonable choice for this assignment.

The most important features are `missed_appointments_last_year`, `previous_advising_visits`, `current_gpa`, `advisor_rating_previous`, and `student_level_First Year`. SHAP and LIME confirm that the model uses both risk-increasing and risk-reducing signals. For example, previous missed appointments and lower GPA can increase Not Completed risk, while no missed appointments can reduce risk for an individual case.

The fairness analysis raises important concerns because accuracy and predicted positive rates differ across age group, gender, and region. Therefore, the final recommendation is that the model is suitable as an early prototype and decision-support tool, but it is not ready for independent real-world deployment. Before operational use, the university should test the model on newer data, validate fairness across groups, review thresholds, and ensure that predictions are used to offer supportive interventions rather than to penalize students.

For the next iteration, the university should collect more appointment records, retest fairness with larger group samples, and compare this interpretable Decision Tree with models such as Random Forest or Gradient Boosting before considering operational use.

18. Limitations

- The dataset is relatively small, with 360 records. This matters especially for cross-validation and fairness analysis because each fold has only about 72 records and some fairness groups in the test set have only 15 to 19 records. In those small groups, one or two changed predictions can noticeably change the percentage results.
- The dataset may not include several advising-context factors that can affect appointment completion, such as student work schedules, transportation reliability, caregiving responsibilities, mental health, financial stress, course workload, or program-specific advising culture. These unmeasured factors may explain some false positives and false negatives.
- The model is a simple Decision Tree. This supports interpretability and is suitable for the assignment, but it may not capture more complex patterns that could be learned by larger or ensemble models such as Random Forest or Gradient Boosting.
- Fairness columns were not used in model training, and `student_age` was removed as an age proxy. However, other variables, such as student level, credits completed, appointment type, or program area, may still indirectly reflect group differences and should be monitored.
- Feature importance, SHAP, and LIME explain model behavior but do not prove that any feature directly causes appointment non-completion. The results should be interpreted as predictive signals, not causal evidence.
- The model should be tested on future advising data and reviewed with advising staff before being used in real business decisions. It should support helpful outreach, not punitive treatment or automatic decisions.

19. Conclusion

This Assignment 6 project successfully completed a full model evaluation, explainability, and fairness reflection workflow for university advising appointment non-completion prediction. The project loaded and inspected the dataset, defined a business-risk target, prepared features through a preprocessing pipeline, trained a Decision Tree model, evaluated performance, interpreted a confusion matrix, analyzed errors, applied cross-validation, checked overfitting, reviewed feature importance, explained predictions with SHAP and LIME, and reflected on fairness across age group, gender, and region.

The model provides useful early insight into appointment non-completion risk, especially because it achieves strong recall for the Not Completed class. However, the model also creates false positives and shows group-level performance differences. Therefore, the model should be treated as a prototype for decision support. It can help advising staff prioritize outreach, but final decisions should involve human judgment, fairness monitoring, and careful business interpretation.

Appendix A: Decision Tree Visualization

The decision tree visualization provides a visual overview of the rules learned by the model. It is included as an appendix because the tree is useful for technical review, while the main business interpretation is provided in the earlier sections.

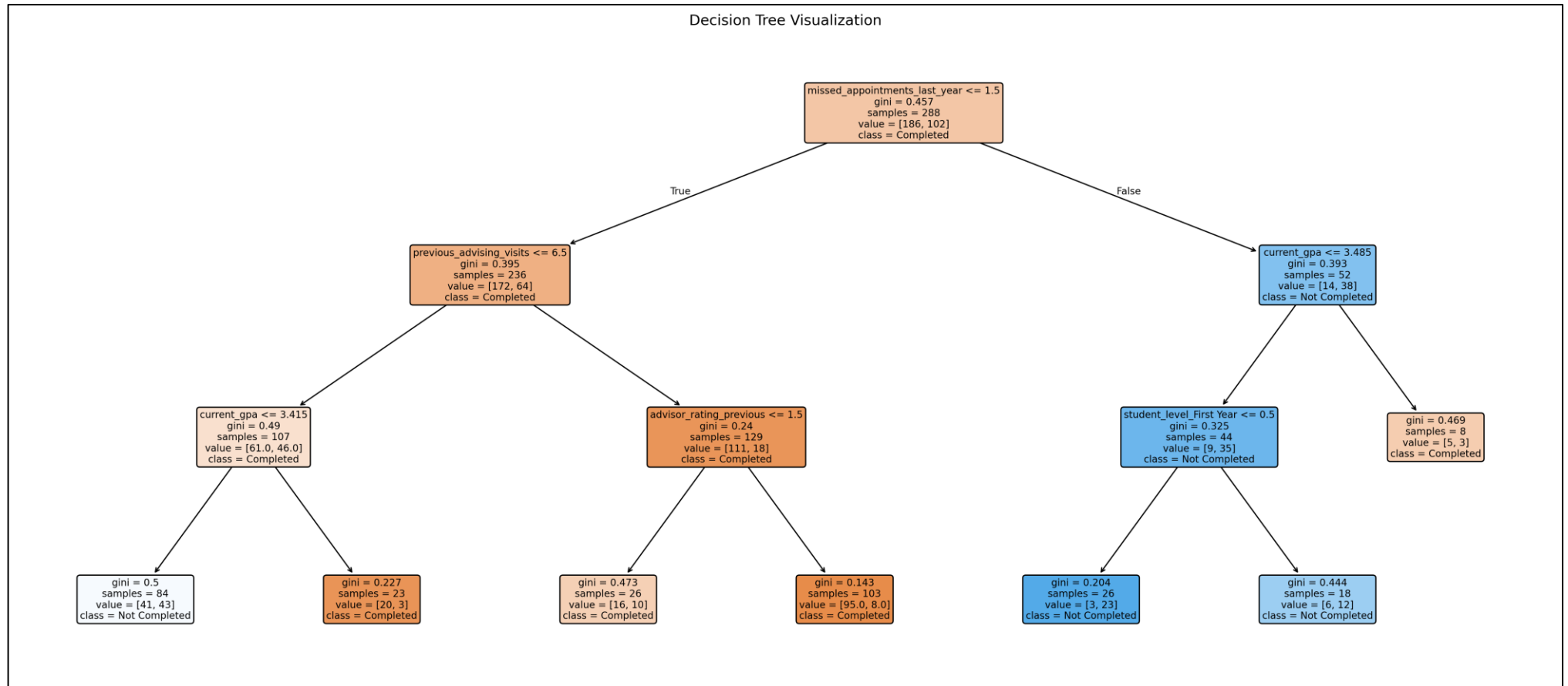


Figure 16: Decision Tree visualization

Appendix B: Assignment Alignment Checklist

Table 29: Assignment alignment checklist

Task	Requirement	How the Report Addresses It
Task 1	Load and understand the dataset	See Section 4, Tables 3-6, and Figures 1-2 for dataset size, columns, data types, missing values, duplicates, and target distribution.
Task 2	Define features and target variable	See Section 5 and Table 8 for target, input features, removed columns, proxy column removal, and fairness-only columns.
Task 3	Data preprocessing	See Section 6 and Table 9 for median imputation, most-frequent imputation, one-hot encoding, and Pipeline use.
Task 4	Train-test split	See Section 7, Tables 10-11, and Figure 3 for the 80/20 stratified split, training/testing sizes, and purpose of split.
Task 5	Train a classification model	See Section 7 and Table 12 for Decision Tree training, predictions, and sample actual-versus-predicted values.
Task 6	Evaluate the model	See Section 8, Tables 13-14, and Figure 4 for accuracy, precision, recall, F1-score, and business meaning.
Task 7	Confusion matrix	See Section 9, Tables 15-16, and Figure 5 for the confusion matrix, TN/FP/FN/TP, and business error meaning.
Task 8	Error analysis	See Section 10, Tables 17-18, and Figure 6 for error type counts, visualization, wrong-prediction sample, and pattern-level discussion.
Task 9	Cross-validation	See Section 11, Table 19, and Figure 7 for 5-fold cross-validation scores, mean, standard deviation, and stability interpretation.
Task 10	Overfitting and underfitting analysis	See Section 12, Tables 20-21, and Figure 8 for training/testing accuracy and tree-depth comparison.
Task 11	Feature importance	See Section 13, Table 22, and Figure 9 for feature importance, top features, visualization, and no-causation caution.
Task 12	SHAP explanation	See Section 14, Table 23, and Figures 10-11 for SHAP summary and individual record interpretation.
Task 13	LIME explanation	See Section 15, Tables 24-25, and Figure 12 for LIME probability, explanation table, and supporting/opposing feature conditions.
Task 14	Fairness and bias reflection	See Section 16, Tables 26-28, and Figures 13-15 for age group, gender, and region fairness summaries and visualizations.
Task 15	Final business interpretation	See Section 17 for final recommendation, business impact, errors, explainability, and fairness concerns.